A DEVELOPER'S DISCOVERY

I Asked AI to Design UI. It Couldn't See What It Built.

Clean code, broken layouts. How I discovered AI's biggest blind spot — and gave it eyes.

Claude Code | AI + UI Design | Playwright MCP | Context Engineering

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THE SETUP

AI Writes Beautiful Code. So What's Wrong?

I've been using Claude Code to build visual outputs — LinkedIn carousels, a VS Code dashboard, brand guidelines.

📄

Carousels

PDF slides with charts & diagrams

💻

Dashboard

React + TypeScript multi-panel layout

🎨

Brand Guide

Color, typography, logo placement

WHAT AI DOES WELL

Understands flexbox, grid, CSS variables. Writes clean HTML. **Technically fluent.**

WHAT AI CAN'T DO

Open a browser. See the rendered page. Spot that text overflowed or columns collapsed. **It writes blind.**

THE FAILURE MODE

Text Everywhere.
Nothing Aligned.

The code looked correct, but the output was broken.

❌ AI'S ATTEMPT

STAT BOXES

0% GPU

14.41% iGPU

58°C Overlaps!

← Text overflows

✅ SHOULD LOOK LIKE

STAT BOXES

0% GPU

14% iGPU

15W Power

58°C Temp

❌ FLOW DIAGRAM

Input → Embed → Search

✅ PROPERLY ALIGNED

Input → Embed → Search

Every one of these issues is **invisible in the code.**
Valid CSS. Semantic HTML. It just doesn't *look* right.

THE PATTERN

Same Mistakes. Every Time.

After dozens of iterations, the same four categories of visual errors kept appearing.

📄

Text Overflow

Long labels spill out of fixed-width containers.

📌

Element Overlap

Absolute positioning without checking neighbors.

📈

Misaligned Diagrams

Flow arrows that don't connect. Boxes at different heights.

🚲

Grid Collapse

4-column grid collapses to one. Side by side becomes stacked.

The frustrating part: fix one issue, ask for another change, and a *different* alignment breaks.

~3–5x	Most	0
More iterations (est.)	Fixes broke more	Visual errors self-caught

THE ROOT CAUSE

A Designer Who Can't Look Up From the Code

The gap isn't intelligence. It's perception.

Human Designer

Write CSS

↓

Look at browser

↓

Adjust

↻

Feedback loop

VS

AI Code Agent

Write CSS

↓

🚫 Can't see output

↓

Hope it works?

❌

No feedback loop

A human designer's superpower isn't better CSS —
it's the ability to **look at the screen and react.**

THE FIX

Give AI Eyes. Playwright MCP.

Playwright MCP lets Claude Code launch a browser, render the page, and screenshot what it built.

✍️ Write HTML/CSS → 🌐 Render page → 📸 See screenshot

↓

👀 Spot overflow → 🔧 Fix the CSS → ✅ Verify fix

```
// .mcp.json
{
  "mcpServers": {
    "playwright": {
      "command": "npx",
      "args": ["@playwright/
        mcp@latest"]
    }
  }
}
```

Same AI, same code generation. Now it can see what it produces and fix it — just like a human would.

THE DIFFERENCE

Context Is Everything.

The model didn't change. The tools around it did.

🚫

Without Playwright

- Write code → hope it works
- Fix one thing → break another
- 3–5 rounds of back-and-forth
- Human becomes the "eyes"
- Tedious and frustrating

✅

With Playwright

- Write → screenshot → verify
- Spots own mistakes first
- Self-correcting loop
- Human reviews final result
- Faster and autonomous

This applies beyond UI

PDFs
Slides, reports, guides

Dashboards
React, data viz, admin

Emails
HTML across clients

The model is rarely the bottleneck. **The context is.**
Give AI the right tools to close the feedback loop,
and it goes from frustrating to genuinely useful.

KEY TAKEAWAYS

What I Learned
Building UI with AI

👁 **AI can code UI. It can't see UI.**
Valid CSS doesn't mean correct layout.
The rendering gap is real.

📄 **Alignment is the failure mode.**
Text overflow, element overlap, misalignment
— the same issues every time.

🔧 **Tools beat prompts.**
No amount of "be careful with alignment"
in a prompt fixes this. A screenshot helps.

💡 **Context engineering > prompt engineering.**
Don't ask AI to be better. Give it the right
tools to close the loop.

Have you used AI to generate visual outputs?

How do you handle the "can't see what it built" problem? 👇